

# Prime-return Euler products: periodic–nonperiodic dichotomy and a clock-renormalization obstruction

Bin Wang

August 2026

## Abstract

For the integral Hénon automorphism  $H(x, y) = (1 - 6x^2 - y, x)$  and an integral point  $P$ , let  $r_p(P)$  be the return period of  $P \bmod p$ . Starting from the exact return–divisibility identity  $p \mid a_n(P) \iff r_p(P) \mid n$ , we construct two zeta objects from the complete marked cycle of  $P \bmod p$ . For a nonperiodic integral orbit, the countable cycle bouquet has finite fixed-point counts in every order and a well-defined formal Artin–Mazur zeta function. Its Dirichlet specialization

$$Z_P(s) = \prod_p (1 - p^{-sr_p(P)})^{-1}$$

converges normally, is holomorphic, and is zero-free on the full half-plane  $\operatorname{Re} s > 0$ . We identify its multiplicative 0/1 coefficients and its logarithmic derivative, whose weights are  $r_p(P) \log p$  on  $p^{jr_p(P)}$ , rather than von Mangoldt weights. The natural cycle-block operator is compact for  $\operatorname{Re} s > 0$  and trace class for  $\operatorname{Re} s > 3$ , where its Fredholm determinant equals  $Z_P(s)^{-1}$ . Conversely, dividing each prime’s clock by  $r_p(P)$  forces the Riemann Euler factor locally but produces an operator in no finite Schatten class. This is a rigorous nonpromotion result: the forced product erases the return lengths instead of recovering a canonical spectral determinant. The paper is an independent post-four-volume theorem edge. It does not alter the RH-1–RH-361 physical frontier, close Gates A–E, identify Riemann zeros, or prove the Riemann Hypothesis.

## 1 Status relative to the four-volume foundation

The RH corpus through RH-361 is preserved in four disjoint, independently auditable volumes covering respectively RH-1–RH-160, RH-161–RH-241, RH-242–RH-281, and RH-282–RH-361 [2]. Those volumes remain the atomic provenance base for the physical determinant program. Its current route coordinate is

`actual_same_clock_unnormalized_head_transport_open,`

with the unproved first leaf

$$D_{4k}(R) = \sum_{2 \leq n < 4k} \frac{|h_{\sigma,n} - s_{k,n}| R^n}{n} \longrightarrow 0.$$

Nothing below estimates this quantity. RH-362 is activated only by the independent-source trigger: it studies arithmetic return cycles across finite fields. The old physical route and all four long-form volumes remain unchanged.

The executable artifact locks the outer four-volume manifest and its independent verification as frozen inputs. The replay covers four volumes, all 361 numbered sources, 73 archive members,

1,548 dependency hashes, and eight result hashes, with zero failures. RH-362 is not included in or used to rewrite that archive.

The source object comes from the exact arithmetic analysis in Wang [4]. The finite-field and strong-divisibility statements used here are theorems, while the reported exponential–gamma return distributions in that source remain finite-range conjectural evidence and are not used. The prime-wheel and infinite-coprime examples of Wang [3] provide a useful comparison: periodic-orbit zeta data can collapse even when symbolic entropy remains positive. We do not identify either symbolic construction with the present marked cycles.

## 2 Integral and modular return data

Fix

$$H(x, y) = (1 - 6x^2 - y, x), \quad H^{-1}(u, v) = (v, 1 - 6v^2 - u).$$

Thus reduction modulo every integer  $M \geq 1$  is a permutation. For  $P \in \mathbb{Z}^2$ , define

$$r_M(P) = \min\{n \geq 1 : H^n(P) \equiv P \pmod{M}\}, \quad r_1(P) = 1,$$

and, writing  $H^n(P) = (x_n, y_n)$ ,

$$a_n(P) = \gcd(|x_n - x_0|, |y_n - y_0|),$$

with  $\gcd(0, 0) = 0$ . The source theorem is

$$M \mid a_n(P) \iff r_M(P) \mid n. \quad (1)$$

For a nonperiodic integral orbit, every  $a_n(P) > 0$ , and the source also proves the strong divisibility identity

$$\gcd(a_m(P), a_n(P)) = a_{\gcd(m, n)}(P).$$

Only (1) is needed for the first new theorem.

**Theorem 2.1** (Exact finite set of low return ranks). *Let  $P \in \mathbb{Z}^2$  have a nonperiodic integral orbit. For every integer  $K \geq 2$ ,*

$$\{p : r_p(P) < K\} = \left\{ p : p \mid \prod_{1 \leq n < K} a_n(P) \right\}.$$

*In particular the set on the left is finite for every  $K$ , and  $r_p(P) \rightarrow \infty$  in the cofinite-prime sense.*

*Proof.* If  $r_p < K$ , take  $n = r_p$  in (1); then  $p \mid a_{r_p}(P)$ . Conversely, if  $p \mid a_n(P)$  for some  $n < K$ , then (1) gives  $r_p \mid n$ , hence  $r_p < K$ . All factors in the product are positive integers, so it has finitely many prime divisors.  $\square$

The concrete point used in the executable artifact is unconditional.

**Lemma 2.2** (Escape of the primary integral seed). *The orbit of  $P_0 = (0, 0)$  is nonperiodic.*

*Proof.* The first coordinate satisfies

$$x_{n+1} = 1 - 6x_n^2 - x_{n-1},$$

and begins

$$x_0 = 0, \quad x_1 = 1, \quad x_2 = -5, \quad x_3 = -150.$$

Suppose  $x_{n-1} = -b_{n-1}$ ,  $x_n = -b_n$ , with  $b_n > b_{n-1} \geq 5$ . Then

$$x_{n+1} = 1 - 6b_n^2 + b_{n-1} < 0, \quad |x_{n+1}| = 6b_n^2 - b_{n-1} - 1 > b_n.$$

The base pair  $b_2 = 5, b_3 = 150$  starts the induction. Hence the absolute values grow strictly and the orbit cannot return to  $P_0$ .  $\square$

### 3 Pointed cycles and a formal Artin–Mazur bouquet

For a prime  $p$ , let

$$\mathcal{O}_{p,P} = \{P, H_p P, \dots, H_p^{r_p-1} P\}$$

be the complete cycle containing  $P \bmod p$ , and let  $U_{p,P}$  be its cyclic permutation matrix. The complete cycle, rather than a single return indicator, is essential.

**Proposition 3.1** (Local marked-cycle determinant). *For every  $n \geq 1$ ,*

$$\mathrm{Tr}(U_{p,P}^n) = r_p \mathbf{1}_{r_p | n}, \quad \det(I - zU_{p,P}) = 1 - z^{r_p}.$$

Consequently the Artin–Mazur zeta function of this finite cycle is

$$\zeta_{\mathcal{O}_{p,P}}(z) = (1 - z^{r_p})^{-1}.$$

*Proof.* The  $n$ -th power of a cyclic shift fixes every basis vector exactly when  $r_p \mid n$ . The eigenvalues of  $U_{p,P}$  are all  $r_p$ -th roots of unity, whose characteristic product gives  $1 - z^{r_p}$ .  $\square$

Define the countable disjoint union

$$\mathcal{X}_P = \bigsqcup_p \mathcal{O}_{p,P}, \quad F_P|_{\mathcal{O}_{p,P}} = H_p.$$

**Theorem 3.2** (Locally finite formal bouquet). *If  $P$  is nonperiodic over  $\mathbb{Z}$ , then every fixed-point set of  $F_P^n$  is finite and*

$$T_n(P) := \# \mathrm{Fix}(F_P^n) = \sum_{p: r_p | n} r_p = \sum_{p | a_n(P)} r_p.$$

The formal Artin–Mazur zeta function is therefore

$$\mathcal{Z}_P(z) = \exp \left( \sum_{n \geq 1} \frac{T_n(P)}{n} z^n \right) = \prod_p (1 - z^{r_p})^{-1} \in \mathbb{Z}[[z]].$$

*Proof.* By theorem 2.1, only finitely many primes have  $r_p \leq n$ . Proposition 3.1 then gives the first sum in (3.2); equation (1) gives the second. Each coefficient of the exponential depends on finitely many fixed point counts, so the usual finite-cycle Euler-product calculation is valid coefficientwise.  $\square$

*Remark 3.3.* The theorem claims a formal power series, not a positive analytic radius, entropy formula, pressure identity, or zeta function of the full maps  $H_p : \mathbb{F}_p^2 \rightarrow \mathbb{F}_p^2$ . It uses one marked cycle from each distinct finite dynamical system.

### 4 The return-rank Dirichlet Euler product

Use the real logarithm of  $p$  and write  $p^{-s} = e^{-s \log p}$ .

**Theorem 4.1** (Zero-free return Euler product). *For a nonperiodic integral point  $P$ ,*

$$Z_P(s) := \prod_p \det(I - p^{-s} U_{p,P})^{-1} = \prod_p (1 - p^{-s r_p})^{-1}$$

converges normally on every closed half-plane  $\mathrm{Re} s \geq \delta > 0$ . It defines a holomorphic, zero-free function on  $\mathrm{Re} s > 0$ , with canonical logarithm

$$\log Z_P(s) = \sum_p \sum_{j \geq 1} \frac{p^{-s j r_p}}{j}.$$

*Proof.* Fix  $\delta > 0$  and choose  $K$  with  $\delta K > 1$ . Theorem 2.1 leaves only finitely many primes with  $r_p < K$ . For all remaining primes and  $\operatorname{Re} s \geq \delta$ ,

$$|p^{-sr_p}| \leq p^{-\delta K}, \quad \sum_p p^{-\delta K} < \infty.$$

After discarding finitely many initial factors, the logarithmic geometric series is uniformly dominated by a constant multiple of this convergent prime sum. Thus (4.1) converges normally. Its exponential is holomorphic and never zero; every finite exceptional factor is also holomorphic and nonzero in the open right half-plane.  $\square$

**Corollary 4.2** (Multiplicative coefficients and trace weights). *The absolutely convergent Dirichlet expansion is*

$$Z_P(s) = \sum_{m \geq 1} \frac{c_P(m)}{m^s}, \quad c_P(m) = \mathbf{1}_{\{r_p(P)|v_p(m) \text{ for every } p\}}.$$

*The function  $c_P$  is multiplicative, takes only the values zero and one, and need not be completely multiplicative. Both its ordinary and absolute convergence abscissae equal zero. Moreover*

$$-\frac{Z'_P}{Z_P}(s) = \sum_p \sum_{j \geq 1} r_p \log p \, p^{-sjr_p}.$$

*Proof.* Expand each local geometric series and use unique factorization; this gives (4.2). Theorem 4.1 gives absolute convergence for every positive real part. At  $s = 0$ , the infinitely many distinct terms  $m = p^{r_p}$  all have coefficient one, so even ordinary convergence fails. Normal convergence permits termwise differentiation on compact subsets, giving (4.2).  $\square$

The support and weight in (4.2) differ from the classical von Mangoldt ledger  $-\zeta'/\zeta = \sum_{p,j} \log p \, p^{-sj}$ . This difference is a theorem, not a missing numerical calibration.

**Theorem 4.3** (No meromorphic crossing at the origin). *Neither  $Z_P$  nor  $D_P = Z_P^{-1}$  admits a meromorphic continuation through  $s = 0$ . This statement does not assert that the full line  $\operatorname{Re} s = 0$  is a natural boundary.*

*Proof.* For real  $\sigma > 0$ ,  $0 < D_P(\sigma) < 1$ . Fix any  $N$  primes  $p_1, \dots, p_N$ . Since  $1 - e^{-x} \leq x$ ,

$$0 < D_P(\sigma) \leq \prod_{j=1}^N (1 - p_j^{-\sigma r_{p_j}}) \leq \sigma^N \prod_{j=1}^N r_{p_j} \log p_j.$$

If  $D_P$  were meromorphic at zero, its decay along the positive real axis would make the singularity removable. The resulting analytic function would have a zero of every order and hence vanish identically, contradicting  $D_P(\sigma) > 0$ . A meromorphic continuation of  $Z_P$  would give one for its reciprocal, so it is also impossible.  $\square$

## 5 Natural cycle blocks and the Fredholm region

Let

$$\mathcal{H}_P = \bigoplus_p \ell^2(\mathcal{O}_{p,P}), \quad T_s = \bigoplus_p p^{-s} U_{p,P}.$$

**Theorem 5.1** (Compact and Schatten regions). *For  $\sigma = \operatorname{Re} s > 0$ ,  $T_s$  is compact. For every  $q > 0$ ,*

$$\|T_s\|_{S_q}^q = \sum_p r_p p^{-q\sigma}.$$

Since  $r_p \leq p^2$ , the condition  $q\sigma > 3$  is sufficient for  $T_s \in S_q$ . In particular, for  $\operatorname{Re} s > 3$ ,

$$\det_F(I - T_s) = \prod_p (1 - p^{-sr_p}) = Z_P(s)^{-1}.$$

If an integer  $m \geq 1$  satisfies  $m \operatorname{Re} s > 3$ , then

$$Z_P(s)^{-1} = \exp\left(-\sum_{n=1}^{m-1} \frac{\tau_n(s)}{n}\right) \det_m(I - T_s), \quad \tau_n(s) = \sum_{p: r_p | n} r_p p^{-sn}.$$

*Proof.* The block norms  $p^{-\sigma}$  tend to zero, so the finite-rank truncations converge to  $T_s$  in operator norm. Each block contributes  $r_p$  equal singular values  $p^{-\sigma}$ , which gives (5.1). The orbit of  $P \bmod p$  lies in a set of  $p^2$  points, so  $r_p \leq p^2$ , and  $\sum_p p^{2-q\sigma} < \infty$  when  $q\sigma > 3$ . The trace-class determinant factors blockwise and proposition 3.1 gives (5.1). The standard definition of the regularized determinant [1], together with the finite combinatorial trace sums  $\tau_n$ , gives (5.1).  $\square$

*Remark 5.2.* Equation (5.1) licenses an ordinary Fredholm interpretation only in the proved trace-class region. The scalar function  $Z_P^{-1}$  continues holomorphically to  $\operatorname{Re} s > 0$ , but this continuation is not by itself a theorem that  $T_s$  is trace class there. Moreover  $T_s$  is an  $s$ -dependent direct sum, uses a marked point, and mixes distinct finite fields. It is not the canonical physical determinant required by Gate A.

## 6 Periodic comparison

The nonperiodic hypothesis is exactly what makes the bouquet locally finite and moves the Euler product's convergence boundary to zero.

**Proposition 6.1** (Periodic branch). *Suppose  $P \in \mathbb{Z}^2$  has exact integral period  $N$ . Then  $r_p(P) \mid N$  for every prime. The exceptional set*

$$E_P = \{p : r_p(P) < N\} = \left\{ p : p \mid \prod_{\substack{d \mid N \\ d < N}} a_d(P) \right\}$$

*is finite, and for  $\operatorname{Re} s > 1/N$ ,*

$$Z_P(s) = \zeta(Ns) \prod_{p \in E_P} \frac{1 - p^{-Ns}}{1 - p^{-r_p(P)s}}.$$

*The right side gives a meromorphic continuation and a simple pole at  $s = 1/N$ . In this branch  $F_P^N$  fixes infinitely many bouquet points, so the formal locally finite Artin–Mazur construction of theorem 3.2 is unavailable. For  $N = 1$ , the proper-divisor product defining  $E_P$  and the finite Euler correction over  $E_P$  are both understood as empty products.*

*Proof.* Integral periodicity gives  $H^N(P) = P$ , hence  $r_p \mid N$ . If  $r_p < N$ , then  $r_p$  is a proper divisor of  $N$  and (1) puts  $p$  among the displayed finite prime divisors. The converse follows in the same way. Outside  $E_P$ , every local factor is  $1 - p^{-Ns}$ , so separating the finite exceptional factors from Euler's product proves (6.1).  $\square$

*Remark 6.2.* The source lock does not assert that this particular Hénon map has an integral periodic point. More importantly, (6.1) imports  $\zeta(Ns)$  because the local periods are almost everywhere equal to  $N$ . It is not an independent spectral reconstruction of zeta or its zeros.

## 7 Why inverse-length weighting is not a bridge

One may force the desired Euler factor by changing every prime's clock:

$$\tilde{T}_s = \bigoplus_p p^{-s/r_p} U_{p,P}.$$

Blockwise,

$$\det(I - p^{-s/r_p} U_{p,P}) = 1 - p^{-s},$$

and therefore the scalar product of inverse block determinants is  $\zeta(s)$  for  $\operatorname{Re} s > 1$ . The output no longer depends on  $r_p$ .

**Theorem 7.1** (Clock-renormalization Schatten obstruction). *Let  $P$  be nonperiodic. For every finite  $q > 0$  and every  $\sigma = \operatorname{Re} s > 0$ ,*

$$\|\tilde{T}_s\|_{S_q}^q = \sum_p r_p p^{-q\sigma/r_p} = \infty.$$

*Thus the forced Riemann Euler product is not the ordinary or finite-order regularized Fredholm determinant of this block direct sum on any positive half-plane.*

*Proof.* By theorem 2.1, eventually  $r_p > 2q\sigma$ . For those primes,  $q\sigma/r_p < 1/2$ , so each summand in (7.1) is at least  $p^{-1/2}$ . The prime sum diverges.  $\square$

This is the decisive scoped negative result. The local normalization  $s \mapsto s/r_p$  prescribes that one complete cycle has weight  $p^{-s}$ . It therefore deletes the very return-length data that the construction was supposed to explain. The exact scalar Euler product is not evidence for a Hilbert–Polya operator, a time-oriented scattering completion, or an intrinsic arithmetic trace.

## 8 Executable protocol and claim firewall

The executable artifact performs only finite exact checks:

1. it hashes the frozen four-volume foundation records together with the return–divisibility and prime-wheel sources;
2. it reproduces the first integral iterates and gcd terms for  $P_0$ ;
3. it verifies (1) for every prime at most 43 and every order at most 12;
4. it checks (2.1) at  $K = 5$ , giving the exact set  $\{2, 3, 5\}$ ;
5. it records complete-cycle traces for selected primes and verifies the finite rank bound  $r_p \leq p^2$  through  $p \leq 43$ ;
6. it requires every Gate and every forbidden macro claim to remain false.

These rows reproduce formulas; they do not prove normal convergence or an all-prime distribution law. The all-prime statements are theorems proved in sections 2–7.

The exact crosswalk is:

| Item      | Status     | Reason                                          |
|-----------|------------|-------------------------------------------------|
| Trigger 1 | untouched  | no actual $D_{4k}(R)$ theorem                   |
| Trigger 2 | untouched  | no typed $q/E_{\text{off}}$ theorem             |
| Trigger 3 | untouched  | no complete unnormalised physical prefix        |
| Trigger 4 | untouched  | no RH-241 noisy envelope/anchor bridge          |
| Trigger 5 | active     | independent source-backed zeta theorem edge     |
| Gates A–E | false/open | no physical determinant-to-completed-zeta chain |

In particular, the paper constructs no Hilbert–Polya operator, identifies no Riemann zero as a spectrum, proves no von Mangoldt-weighted prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

## 9 Conclusion

The return–divisibility theorem supports a natural all-prime zeta object, but its analytic behavior is sharply different from the Riemann zeta function on the nonperiodic branch: the return Euler product is already holomorphic and zero-free on the whole open right half-plane. The natural block operator retains the return ranks and has a genuine, though restricted, Fredholm region. The inverse-length clock produces the desired Riemann local factor only by making every cycle contribute the same prescribed prime weight, and the resulting direct sum loses all finite Schatten classes.

The narrow next arithmetic questions are therefore source-backed constraints on the ranks themselves—average-over-primes theorems, Chebotarev-type finite-period events, primitive divisors, or height growth—and a genuinely intrinsic global dynamics whose determinant does not arise from independently retiming every prime. The original physical RH route remains available and unchanged on top of the four-volume foundation.

## References

- [1] Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, second edition, 2005. doi: 10.1090/surv/120.
- [2] Bin Wang. Conditional prime dynamics: A corpus frontier synthesis. Repository synthesis of RH-1–RH-361, 2026.
- [3] Liang Wang. Periodic-orbit collapse in B-admissible shifts: Exact prime-wheel zeta functions and infinite coprime limits. Repository preprint, commit 7fd3a3fdd5a6a25827a0965345459baf4a47b816, 2026. URL [https://github.com/maris205/dyna\\_zeta\\_map/tree/7fd3a3fdd5a6a25827a0965345459baf4a47b816](https://github.com/maris205/dyna_zeta_map/tree/7fd3a3fdd5a6a25827a0965345459baf4a47b816).
- [4] Liang Wang. Prime-sampled return times for an integral Hénon map. Repository preprint, commit c37d191672d30de49b2054be3a03cf2db068694f, 2026. URL [https://github.com/maris205/henon\\_prime\\_returns/tree/c37d191672d30de49b2054be3a03cf2db068694f](https://github.com/maris205/henon_prime_returns/tree/c37d191672d30de49b2054be3a03cf2db068694f).